Gevra Opencast Mine

🏭 Introduction

The **Gevra Opencast Mine** is the **largest coal mine in India** and one of the largest opencast mines in Asia. It is operated by South Eastern Coalfields Limited, a subsidiary of Coal India Limited.

The mine is a flagship project in India's coal industry, with an annual production capacity exceeding **50 million tonnes**. It primarily produces **non-coking thermal coal** used for power generation and plays a vital role in meeting the country's energy demands. Due to its massive scale and high level of mechanization, Gevra is often considered a model for large-scale surface mining operations.

🌍 Geographical Overview – Gevra Opencast Mine

The Gevra mine is located in the **Korba district** of Chhattisgarh, within the Korba coalfield, one of the most productive coal regions in India.

📍 Location & Setting

- Situated near **Gevra town in Korba district**
- Part of the **Hasdeo–Arand basin**
- Well connected by rail (Gevra Road) and road networks

🌄 Physical & Geological Features

- Located in a **plateau region with gently undulating terrain**
- Contains **thick and laterally extensive coal seams**, ideal for opencast mining
- Coal type: **non-coking thermal coal**

🌊 Nearby Natural Features

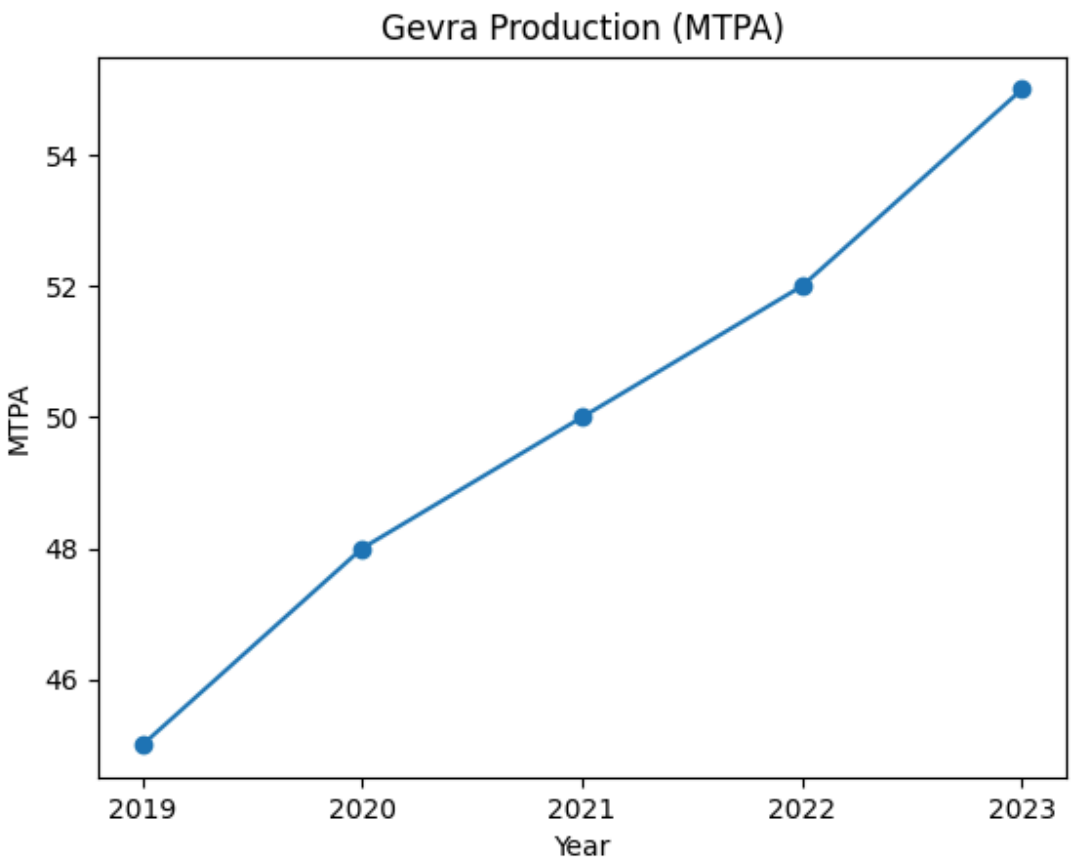
- Located near the Hasdeo River basin
- Close to major thermal power plants in the Korba region

🌦️ Climate

- Tropical climate
- Very hot summers (up to ~47°C)
- Moderate to heavy monsoon rainfall

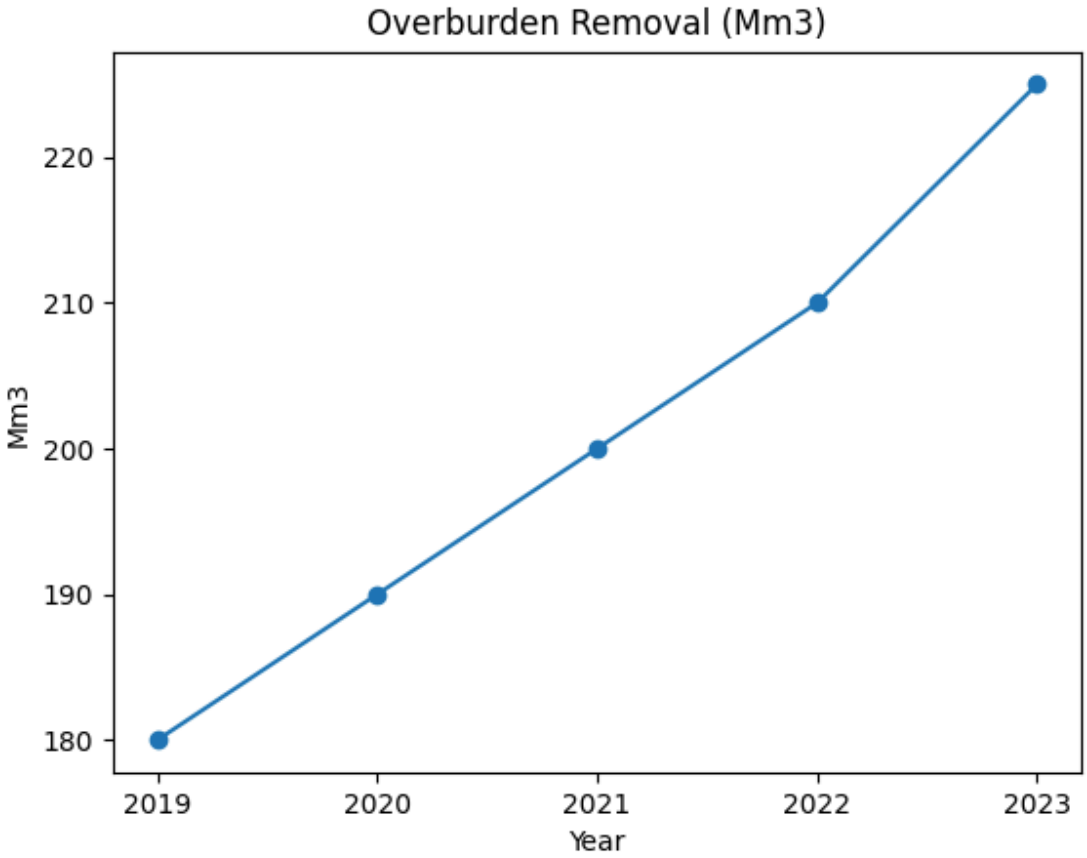
- Mild winters

📊 1. Production Trend (MTPA)



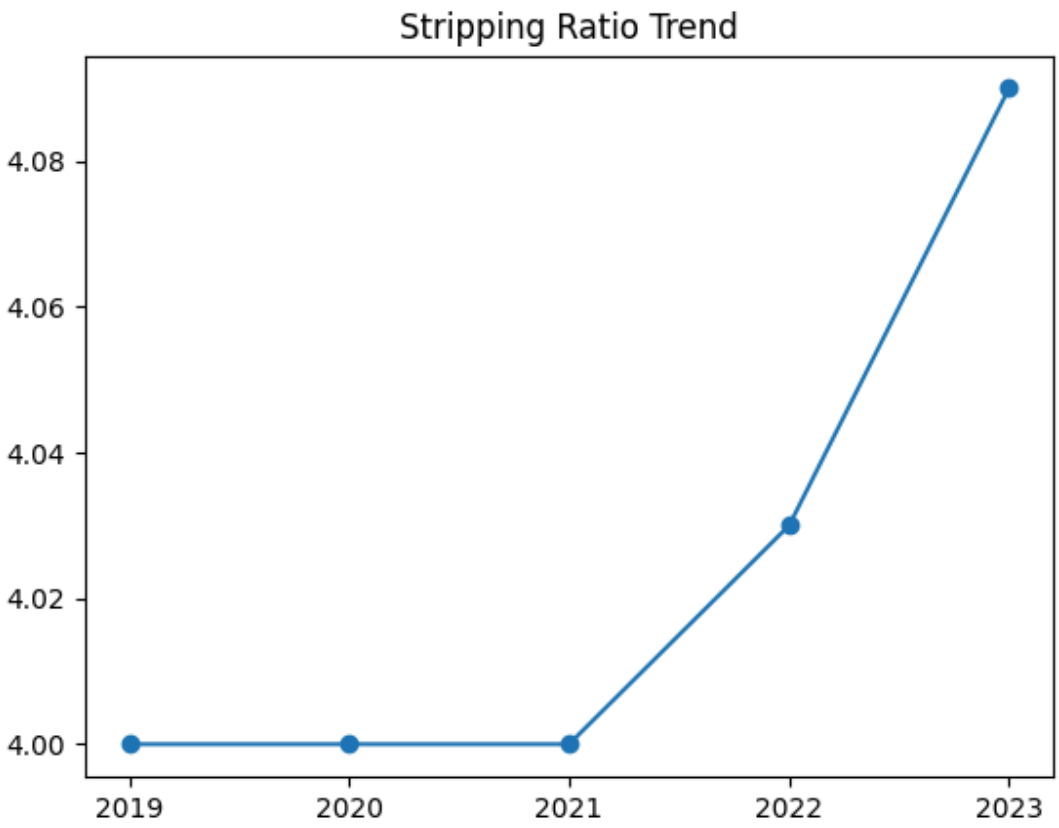
This chart shows the year-wise coal production of Gevra, operated by South Eastern Coalfields Limited.
The increasing trend reflects expansion and high efficiency of large-scale operations.
Significance: Useful for production planning, forecasting, and evaluating mine performance at a national scale.

📊 2. Overburden Removal (Mm³)



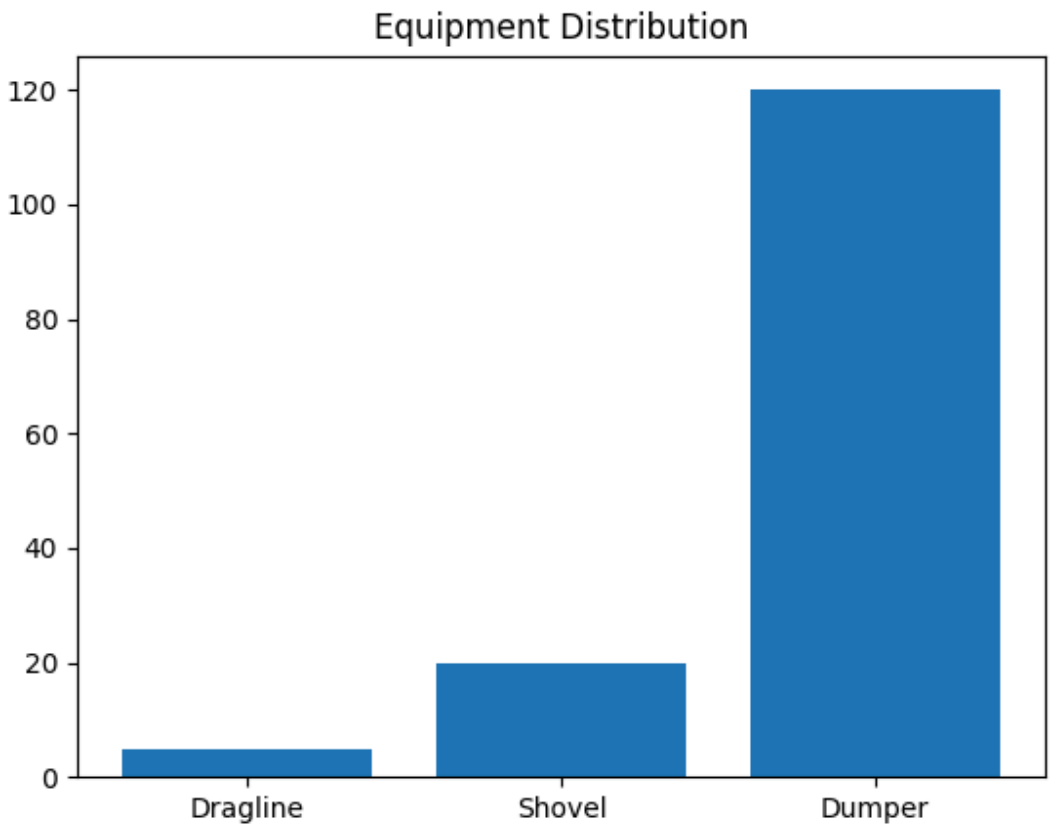
Represents the volume of overburden removed to expose coal seams.
Large values indicate the massive scale of opencast operations at Gevra.
Significance: Essential for calculating stripping ratio and planning dump management and equipment.

📊 3. Stripping Ratio Trend



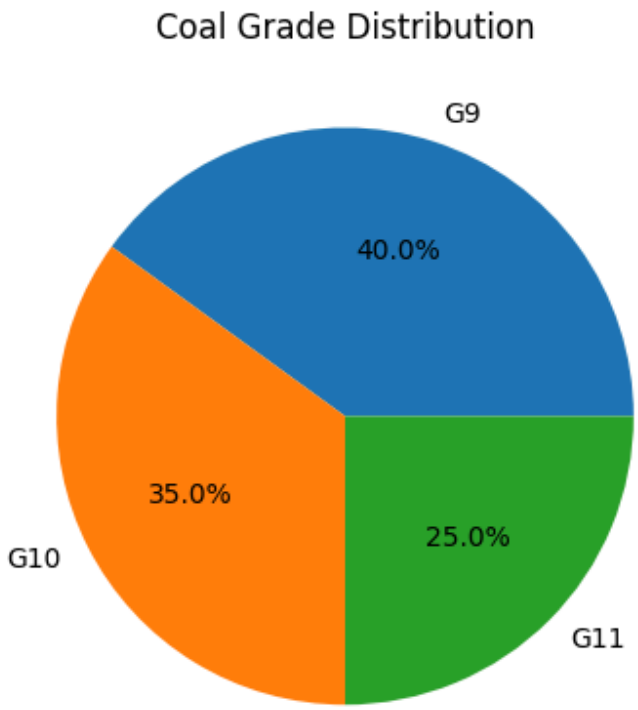
Shows the ratio of overburden to coal production over time.
A relatively stable trend indicates efficient mine planning.
Significance: Key economic parameter affecting cost, productivity, and mine feasibility.

📊 4. Equipment Distribution



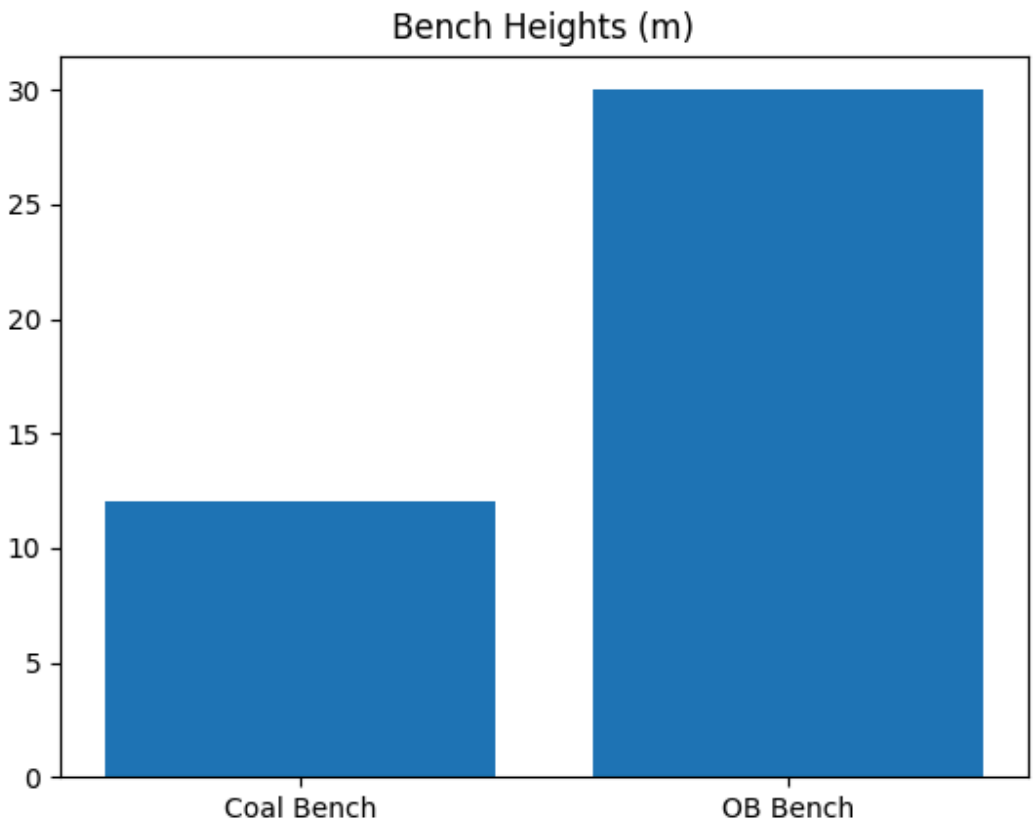
Displays the number of major mining equipment like draglines, shovels, and dumpers.
Dumpers dominate due to high material movement requirements.
Significance: Helps in fleet optimization, productivity analysis, and equipment planning.

📊 5. Coal Grade Distribution



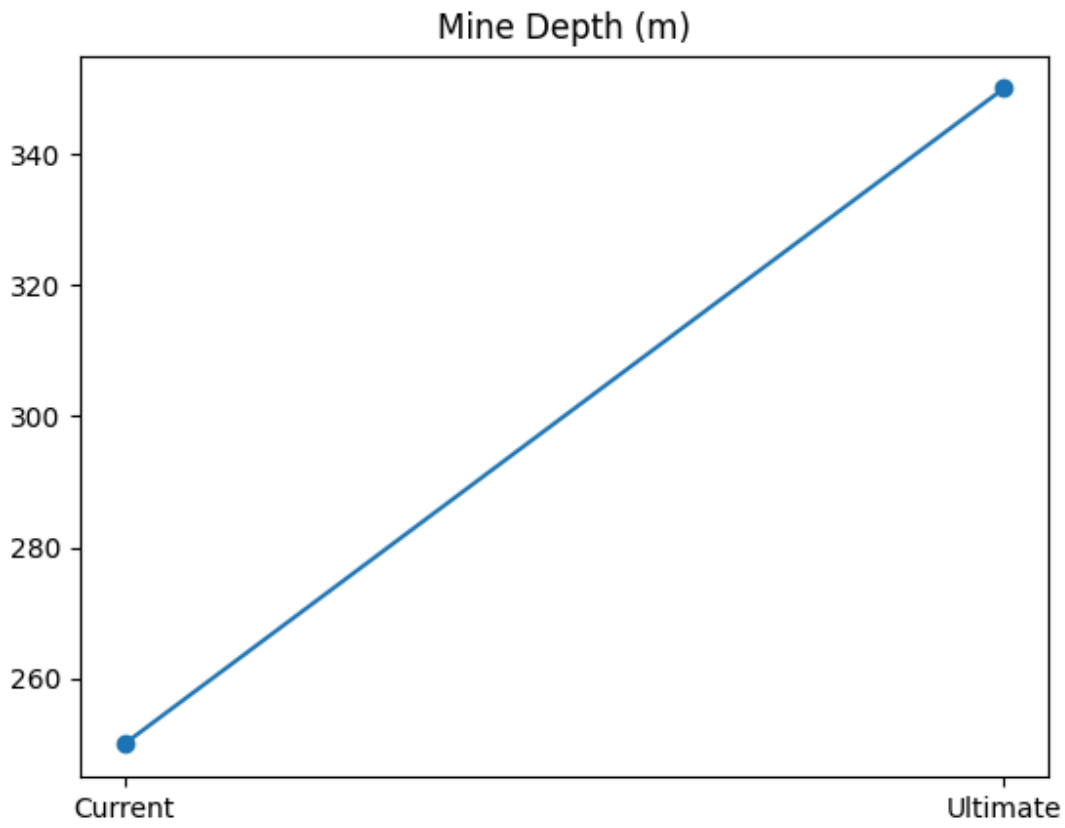
Shows the proportion of different non-coking coal grades (G9–G11).
These grades are suitable for thermal power generation.
Significance: Important for determining coal quality, pricing, and end-use suitability.

📊 6. Bench Heights (m)



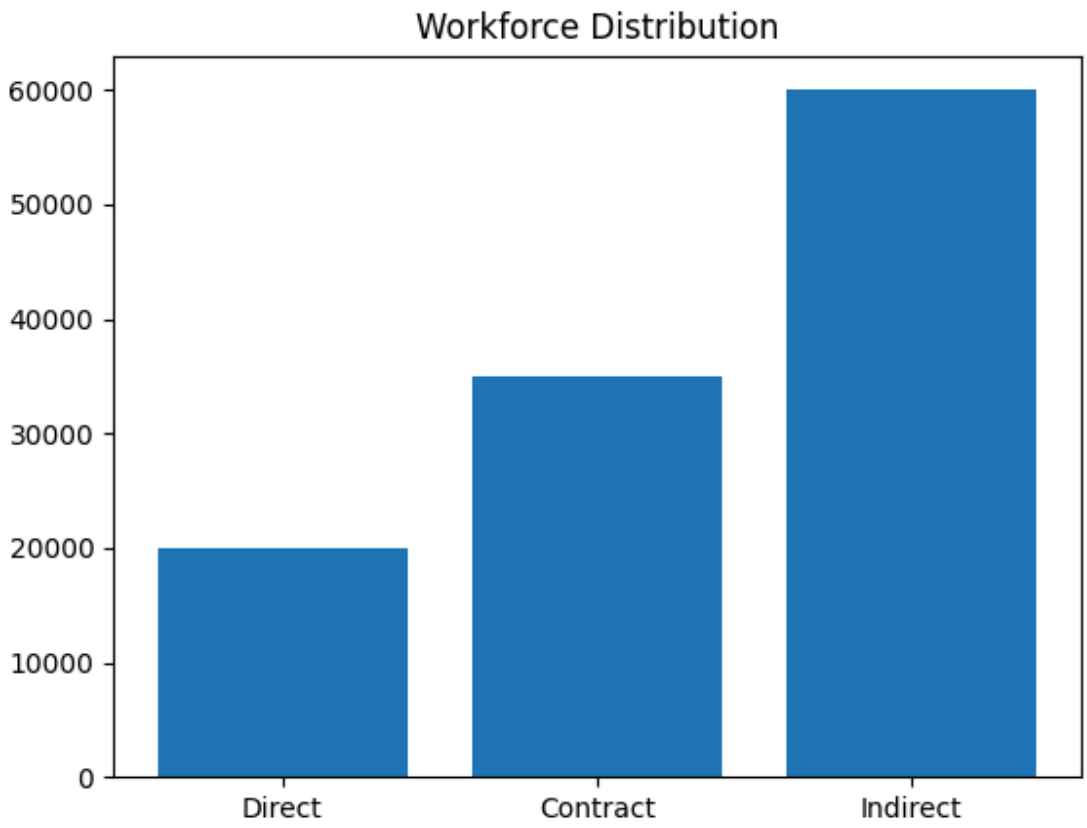
Represents the height of benches used in coal and overburden extraction.
Proper bench design ensures efficient excavation and operational safety.
Significance: Critical for slope stability, equipment compatibility, and safe mining practices.

📊 7. Mine Depth Progression



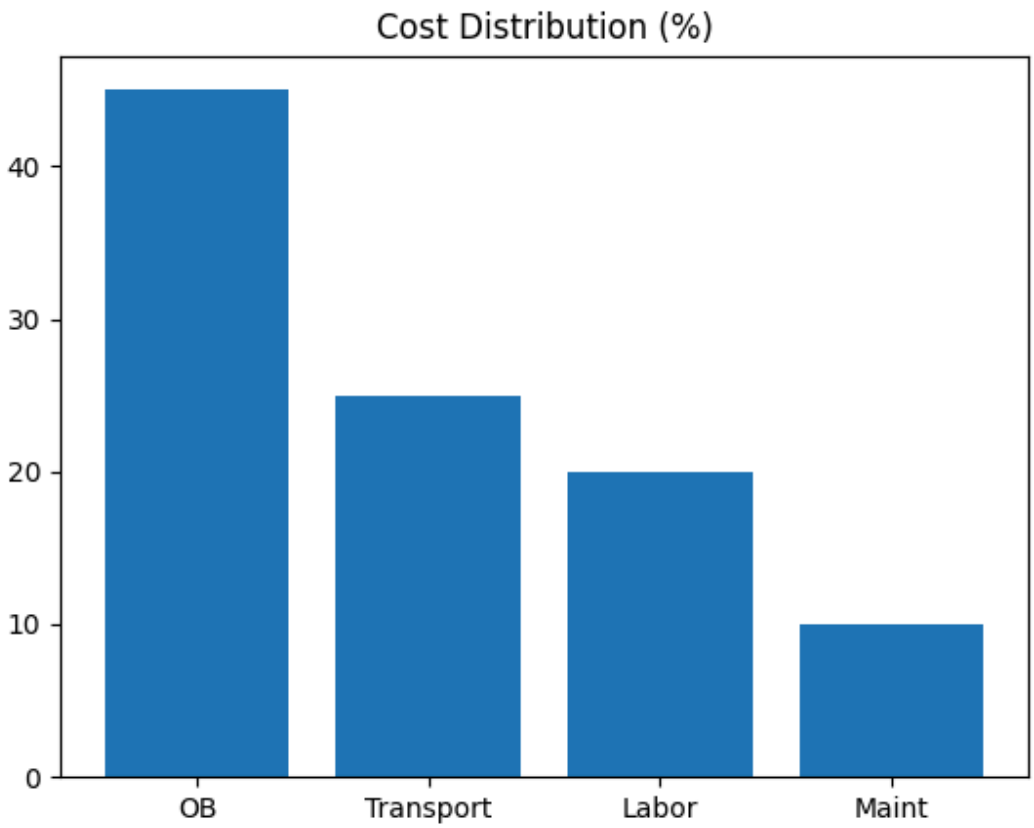
Shows current and ultimate depth of the mine.
Indicates expansion and long-term mining potential.
Significance: Important for mine planning, slope design, and groundwater management.

📊 8. Workforce Distribution



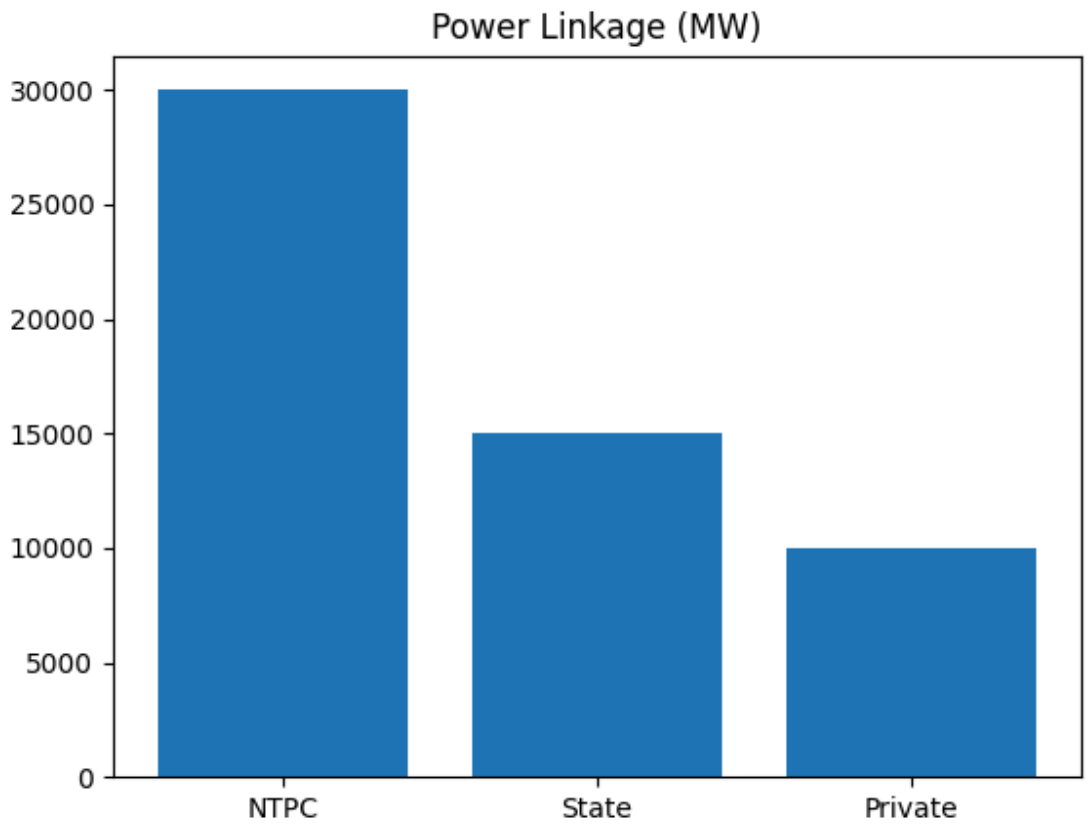
Displays distribution of manpower across direct, contract, and indirect categories.
Reflects the large workforce required for such a massive mine.
Significance: Useful for manpower planning and understanding socio-economic impact.

📊 9. Cost Distribution (%)



Shows breakdown of mining costs into OB removal, transport, labor, and maintenance.
OB removal constitutes the largest share due to scale.
Significance: Helps in cost control, budgeting, and improving operational efficiency.

📊 10. Power Linkage Capacity (MW)



Represents coal supply linkage to different power sectors.
Gevra supplies coal to major power plants across India.
Significance: Highlights the mine's critical role in ensuring national energy security.